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# Luminal Fat Digestion and Absorption

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**Authors:** Ryan Downey, Lance Bridges
**Editors:** Jordan Finn, Russell Johnson
**Contributors:** Ian Murray, Eleanor Schrems, Kevin Kunkler

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**Learning Objectives (5)**

***After completing this brick, you will be able to:***

1. Predict how bile salt deficiency impairs lipid solubilization, and describe the amphipathic basis for emulsification and micelle formation.
2. Analyze the cooperative roles of pancreatic lipase and colipase in triglyceride h onsequences of lipase deficie
3. Trace the intracellular pathway of lipid uptake, re-esterification, and chylomicron assembly within the enterocyte.
4. Distinguish the lymphatic transport route of dietary fat from the portal route of carbohydrates and amino acids, and explain the structural basis for this divergence.
5. Predict the consequences of ileal dysfunction on bile salt pool size and luminal fat digestion.

**CASE CONNECTION**

MS, a 34-year-old male, presents to a gastroenterology clinic with unexplained weight loss and bulky, oily stools over the past month.

What processes might account for MS's inability to absorb dietary fat? How would you distinguish between luminal and absorptive causes of fat malabsorption? Consider your answers as you read, and we'll revisit MS at the end of the brick.

GO TO CONCLUSION ↓

## Bile Salt Emulsification, Micelle Formation, and Deficiency (LO1)

Dietary fat arrives in the small intestine as large, lipid droplets that are immiscible in the aqueous intestinal environment. Before pancreatic enzymes can act efficiently an products, these droplets must be broken into smaller particles and solubilized. Bile salts accomplish both tasks: acting first as emulsifiers to increase the surface area available for enzymatic hydrolysis, and then as the structural backbone of mixed micelles that carry lipid products to the absorptive epithelium.

### Bile Salt Structure

Bile salts have an amphipathic structure (established in the Exocrine Functions brick) — one face hydrophilic, one face hydrophobic. At the oil-water interface, they orient with their hydrophobic face inserted into lipid droplets and their hydrophilic face exposed to the aqueous lumen, lowering surface tension. Mechanical agitation from intestinal peristalsis then breaks large fat globules into smaller droplets — **emulsification** — dramatically increasing the surface area accessible to pancreatic lipase.

### Micelle Formation

Once pancreatic lipase has hydrolyzed triglycerides into free fatty acids (FFAs) and 2-monoglycerides (2-MGs), these products must be transported across the unstirred water layer overlying the enterocyte brush border. Bile salts accomplish this by forming mixed micelles—small, spherical aggregates in which the hydrophobic lipid products are sequestered in the interior while the hydrophilic portions of bile salts face the aqueous lumen.

Micelle formation occurs above a threshold concentration known as the critical micellar concentration (CMC). Mixed micelles typically incorporate FFAs, 2-MGs, cholesterol, fat-soluble vitamins (A, D, E, K), and lysophospholipids. Because micelles are water-soluble particles roughly 3–10 nm in diameter, they diffuse efficiently thr deliver their lipid cargo directly to the

**CLINICAL REASONING**

Fatty acid chain length determines the absorption pathway. Short-chain (SCFAs, ≤5 carbons) and medium-chain fatty acids (MCFAs, 6–12 carbons) — common in dairy fats and released by lingual and gastric lipase — are water-soluble enough to diffuse directly into the portal circulation without micelle formation or chylomicron packaging. Long-chain fatty acids (LCFAs, ≥13 carbons), produced by pancreatic lipase in the small intestine, require the full digestive cascade: bile salt micelles, enterocyte uptake, re-esterification, and chylomicron-mediated lymphatic transport.

This distinction has clinical leverage. When the LCFA pathway fails — through pancreatic insufficiency, bile salt deficiency, ileal dysfunction, or chylomicron assembly defects — substituting dietary fat with MCT (medium-chain triglyceride) supplements bypasses the defect, preserving caloric intake. **Serum lipase and amylase** remain the standard biomarkers of pancreatic injury.

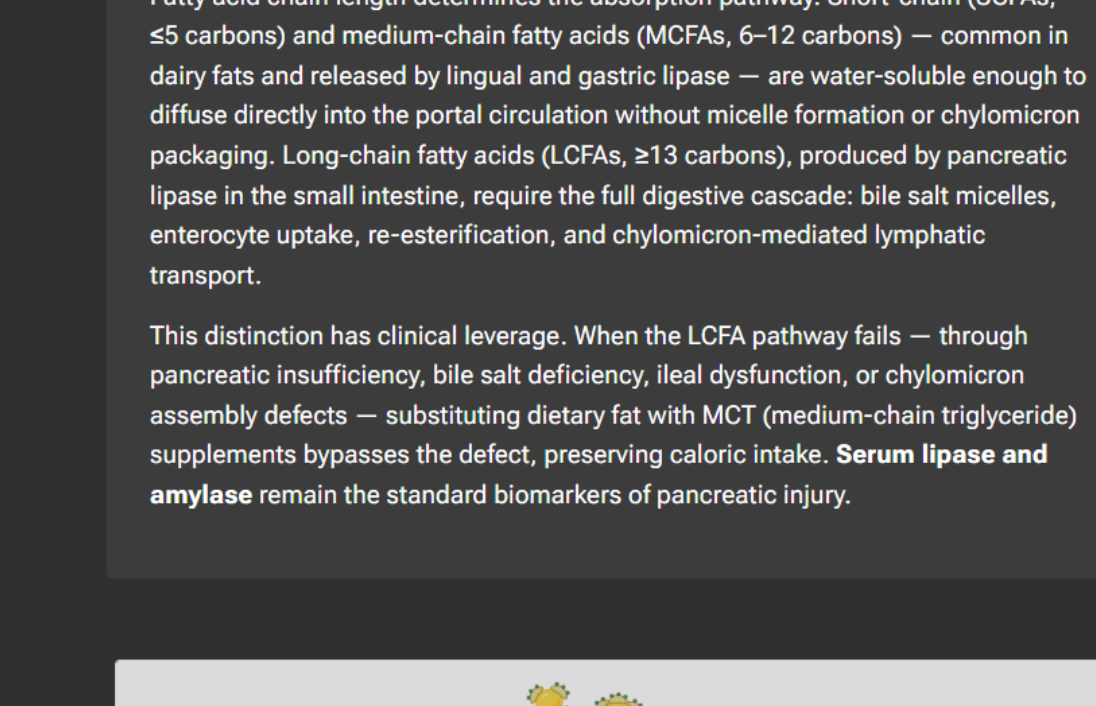
Fat droplets
Fatty acids and 2-monoglycerides
Bile salt emulsified fat droplets carry lipid products to the small intestine.
Pancreatic lipase hydrolyzes triglycerides into free fatty acids and 2-monoglycerides.
Within mixed micelles, these products reach the brush borde and are taken up by the enterocyte.
Once in the cell, fatty acids and 2-monoglycerides are re-esterified into triglycerides, packaged into chylomicrons, and secreted into the lacteal.
Re-esterification
Triglyceride
Small intestine lumen
Small intestine epithelium
Villus capillaries
Lacteal

**Figure 1: Bile salts deliver the products of fat digestion to the enterocyte for re-esterification, chylomicron packaging, and lymphatic transport via thelacteal.**

### Effects of Bile Salt Deficiency

When the intraluminal concentration of bile salts falls below the CMC—due to liver disease, biliary obstruction, or ileal dysfunction—micelle formation fails and emulsification efficiency is reduced. Lipid droplets remain large, limiting the surface area accessible to pancreatic lipase.

The clinical consequence is steatorrhea: bulky, greasy stools rich in unabsorbed fat. Fat-soluble vitamin deficiencies (A, D, E, K) develop over time, leading to night blindness, osteomalacia, neuropathy, and coagulopathy, respectively.

What is the pr fat digestion?

## Pancreatic Lipase, Colipase, and Triglyceride Hydrolysis (LO2)

Dietary triglycerides (TGs) account for over 90% of ingested fat. Although lingual and gastric lipases initiate some hydrolysis in the stomach, the vast majority of TG digestion occurs in the duodenum and jejunum under the action of pancreatic lipase.

### Lipase–Catalyzed Breakdown

Pancreatic lipase cleaves ester bonds at the sn-1 and sn-3 positions of TGs, generating two FFAs and one 2-MG per TG molecule. The enzyme is water-soluble and must act at the oil-water interface of emulsified lipid droplets. However, bile salts—while essential for emulsification—paradoxically displace lipase from the droplet surface when present at high concentrations. This is where colipase becomes indispensable.

Colipase is a small protein co-secreted by the pancreas in an inactive proform (procolipase) that is activated by trypsin in the duodenum. Colipase anchors to the lipid droplet surface and provides a binding platform that stabilizes pancreatic lipase at th sence of bile salts, restoring lipase activi

Pancreatic phospholipase A2 is secreted as an inactive proenzyme and activated by trypsin in the duodenal lumen. It acts on biliary phosphatidylcholine—the major phospholipid secreted into bile—cleaving the sn-2 ester bond to yield a lysophospholipid and a free fatty acid. Both products are readily incorporated into mixed micelles: lysophospholipids are amphipathic and contribute to micelle stability, while the released fatty acid joins the micellar lipid core. This is why lysophospholipids appear prominently in the micelle composition: they originate from phospholipase A2 action on biliary phospholipid rather than from dietary phospholipid digestion alone.

### Deficiency Consequences

Pancreatic exocrine insufficiency, as seen in chronic pancreatitis, cystic fibrosis, or pancreatic cancer, reduces lipase and colipase secretion. Without adequate enzyme activity, TGs pass largely unhydrolyzed into the colon, where bacterial fermentation produces gas and osmotic effects. The result is steatorrhea, caloric malnutrition, and fat-soluble vitamin deficiency. This represents a luminal (pre-absorptive) cause of fat malabsorption: digestion fails before lipids reach the enterocyte, which remains structurally intact.

**CLINICAL CORRELATION**

**CYSTIC FIBROSIS**

Cystic fibrosis is the most common inherited cause of pancreatic exocrine insufficiency. The CFTR defect, established in the Exocrine Functions brick, produces viscous pa that block lipase delivery to the duode lipase–bile salt–chylomicron pathway — are the most affected; SCFAs and MCFAs are spared because they bypass this pathway entirely (see Exocrine Functions of the GI System, Cystic Fibrosis Clinical Reasoning).

Management combines pancreatic enzyme replacement therapy with MCT supplementation, which delivers calories through the spared portal route. **CFTR modulators** (e.g., **ivacaftor, elexacaftor**) restore partial channel function in patients with responsive mutations.

**ACTIVE LEARNING QUESTION**

**Pancreatic lipase is displaced from lipid droplet surfaces by bile salts. Which molecule restores lipase activity at the interface?**

- **A.** ○ Cholecystokinin
- **B.** ○ Colipase
- **C.** ○ Phospholipase A2
- **D.** ○ Enteropeptidase
- **E.** ○ Procolipase

Explanation (requires correct answer)

## Enterocyte Lipid Uptake, Re–Esterification, and Chylomicron Assembly (LO3)

Once mixed micelles deliver FFAs and 2-MGs to the apical surface of enterocytes, the lipid products must cross the brush border membrane, be re-assembled into TGs inside the cell, and then be packaged into transport particles for export.

### Lipid Uptake Mechanism

Long-chain FFAs and 2-MGs cross the apical membrane of enterocytes primarily by passive diffusion, driven by the concentration gradient maintained by rapid intracellular esterification. However, a protein-mediated component also exists: fatty acid transport protein 4 (FATP4) and the scavenger receptor CD36 facilitate uptake of long-chain fatty acids, particularly at lower luminal concentrations. Short- and medium-chain fatty acids, being more water-soluble, diffuse freely and bypass chylomicron assembly entirely, passing directly into the portal circulation.

### Cholesterol Absorption

Dietary cholesterol esters must first be hydrolyzed to free cholesterol by pancreatic cholesterol esterase (also called bile salt–stimulated lipase). Free cholesterol is then taken up across the apical membrane via Niemann-Pick C1-like 1 protein (NPC1L1), expressed on the enterocyte brush border. NPC1L1 also mediates reabsorption of biliary cholesterol recycled through the enterohepatic circula sterol is re-esterified by ACAT2 (acyl-CoA cholesterol acyltransferase 2) and incorporated into chylomicrons alongside TGs. NPC1L1 is the direct molecular target of ezetimibe, which competitively inhibits cholesterol uptake and lowers circulating LDL.

**CLINICAL CORRELATION**

**EZETIMIBE AND NPC1L1**

Ezetimibe competitively inhibits NPC1L1, the enterocyte transporter responsible for both dietary and biliary cholesterol uptake. The resulting reduction in cholesterol delivery to the liver triggers compensatory upregulation of hepatic LDL receptors, lowering circulating LDL-C. The effect is additive with statins, which act through a separate mechanism: inhibition of HMG-CoA reductase to reduce hepatic cholesterol synthesis. Ezetimibe is cholesterol-selective and does not meaningfully impair triglyceride, bile salt, or fat-soluble vitamin absorption.

### Fat–soluble Vitamin Absorption

Fat-soluble vitamins (A, D, E, K) are absorbed by a mechanism that mirrors long-chain fatty acid uptake: they are incorporated into mixed micelles in the lumen, delivered to the apical membrane, and cross the brush border by passive diffusion in a process that is entirely bile salt– and micelle-dependent. Once inside the enterocyte, they are incorporated into chylomicrons and exported via the lymphatic route alongside dietary TGs. Because their absorption depends on intact micelle formation, any condition that depletes the bile salt pool below the CMC alongside fat malabsorption.

**CLINICAL CORRELATION**

**CELIAC DISEASE**

Celiac disease is an immune-mediated reaction to dietary gluten in genetically susceptible individuals (HLA-DQ2 and HLA-DQ8). Antibodies against tissue transglutaminase damage the duodenal and jejunal mucosa, producing villous atrophy with crypt hyperplasia and markedly reduced brush border enzyme expression. Luminal digestion remains intact — bile salts, pancreatic lipase, and colipase are all secreted normally — but the absorptive surface is lost, so fat, iron, and folate malabsorption develop despite normal digestion.

Diagnosis relies on serologic testing (anti-tissue transglutaminase IgA) and confirmatory small bowel biopsy showing the characteristic villous atrophy. Treatment is lifelong gluten avoidance. Celiac disease is the prototypical **absorptive (post-luminal)** cause of fat malabsorption, the diagnostic counterpoint to pancreatic insufficiency and bile salt deficiency.

### Re–Esterification Process

Inside the enterocyte, FFAs are activated to fatty acyl-CoA thioesters by acyl-CoA synthetase in the smooth endoplasmic reticulum (SER). The predominant pathway for TG re-synthesis is the monoglyceride pathway: 2-MG is sequentially acylated by monoacylglycerol acyltransferase (MGAT) and diacylglycerol acyltransferase (DGAT) to form diacylglycerol (DAG) and then TG. This pathway use digestion, as the glycerol backbone, bypassing the need for de novo glycerol synthesis.

### Chylomicron Assembly

In the smooth endoplasmic reticulum, microsomal triglyceride transfer protein (MTP) loads newly synthesized TGs, cholesterol esters, phospholipids, and fat-soluble vitamins onto apolipoprotein B-48 (apoB-48), forming a primordial lipoprotein particle that is further processed in the Golgi to form nascent chylomicrons.

ApoB-48 is synthesized exclusively in intestinal enterocytes and is essential for chylomicron assembly; its absence (as in abetalipoproteinemia) leads to complete failure of chylomicron formation and severe fat malabsorption. The assembled chylomicrons are then secreted by exocytosis from the basolateral membrane into the lamina propria.

Chylomicrons are the largest of the lipoprotein particles, with a diameter of 75–1200 nm. Their core is composed predominantly of TGs (80–85%), with smaller amounts of cholesterol esters. The surface monolayer contains phospholipids, free cholesterol, and apolipoproteins (apoB-48, apoA-I, apoA-IV).

**CLINICAL CORRELATION**

**ABETALIPOPROTEINEMIA**

Abetalipoproteinemia is an autosomal recessive disorder caused by mutations in **microsomal triglyceride transfer protein (MTP)**, the chaperone that loads lipids onto apoB-48 in enterocytes and apoB-100 in hepatocytes. Without functional MTP, apoB cannot b micron and VLDL assembly both fail. The clinical consequences trace directly to the lost transport routes: dietary fat accumulates in enterocytes and is lost as steatorrhea, and circulating LDL is markedly reduced.

Without intact chylomicron transport, fat-soluble vitamins (A, D, E, K) cannot be exported to peripheral tissues. **Vitamin E deficiency** dominates the clinical picture, producing progressive neurologic dysfunction (ataxia, areflexia, peripheral neuropathy) from oxidative damage to neuronal membranes. **Acanthocytes** — spiky red blood cells from defective membrane lipid composition — are a classic peripheral smear finding. Treatment is high-dose fat-soluble vitamin supplementation, especially vitamin E.

What is the primary role of chylomicrons, and which apolipoprotein is essential for their assembly?

## Lymphatic and Portal Transport of Absorbed Nutrients (LO4)

After absorption, dietary macronutrients diverge into two distinct transport routes: carbohydrates and amino acids enter the portal circulation and travel directly to the liver, while dietary fat packaged as chylomicrons enters the lymphatic system. This separation is not arbitrary—it reflects fundamental differences in the size, solubility, and metabolic handling of these nutrients.

### Lymphatic Route Details

Chylomicrons secreted from the basolateral surface of enterocytes are too large (75–1200 nm) to enter intestinal capillaries: the villous capillary endothelium is fenestrated, but fenestrae are only ~60–80 nm in diameter, far too small to accommodate chylomicrons. Instead, they enter specialized lymphatic capillaries called lacteals, located in the core of each intestinal villus. Lacteals have discontinuous endothelium and large intercellular gaps that readily accommodate chylomicron-sized particles.

From the lacteals, chylomicron-rich lymph (called chyle) flows through the mesenteric lymphatics into the cisterna chyli and then up the thoracic duct, emptying into the systemic circulation at the junction of the left subclavian and internal jugular veins. This route bypasses the liver on the first pass, allowing dietary fat to be distributed to peripheral tissues, particularly adipose tissue and muscle, before hepatic processing.

### Portal Route Comparison

Monosaccharides (glucose, fructose, galactose) and amino acids are small, water-soluble molecules that are absorbed via specific transporters on the apical and basolatera ss directly into the intestinal capillaries ing to the liver for immediate metabolic processing—glycogen synthesis, gluconeogenesis, protein synthesis, and amino acid catabolism. Short- and medium-chain fatty acids, being water-soluble enough to remain unesterified, also follow this portal route.

### Physiological Rationale

Portal delivery of glucose and amino acids allows the liver to act as a first-pass metabolic gatekeeper, buffering postprandial nutrient surges before they reach systemic circulation. In contrast, lymphatic delivery of chylomicrons allows dietary fat to reach peripheral tissues rapidly, supporting energy needs in muscle and fat storage in adipose tissue, while avoiding immediate hepatic TG extraction that could overwhelm hepatic lipid metabolism.

This distinction also has clinical relevance: patients with intestinal lymphangiectasia (dilated, leaky lacteals) lose chylomicrons and fat-soluble vitamins into the gut lumen, causing protein-losing enteropathy and fat malabsorption despite intact digestion and enterocyte function.

**ACTIVE LEARNING QUESTION**

**Why do chylomicrons enter lacteals rather than intestinal capillaries after secretion from enterocytes?**

- **A.** ○ They are water-soluble
- **B.** ○ ApoB-48 directs them to lymphatics
- **C.** ○ Large size prevents capillary entry
- **D.** ○ Portal vein pressure excludes them
- **E.** ○ They carry fat-soluble vitamins

Explanation (requires correct answer)

## Enterohepatic Bile Salt Recycling and Ileal Dysfunction (LO5)

Bile salt availability in the lumen depends on intact enterohepatic recycling (established in the Exocrine Functions brick). With each meal, the bile salt pool — too small to support digestion through synthesis alone — is sustained by efficient reabsorption in the terminal ileum and hepatic re-secretion. When this recycling fails, particularly at the ileal site of active reabsorption, the consequences for fat digestion are direct and predictable.

### Impact of Ileal Dysfunction

The terminal ileum is the exclusive site of active bile salt reabsorption. When ileal function is compromised—by Crohn's disease, surgical resection, or radiation enteritis—the ASBT-mediated reabsorption fails, and a larger fraction of bile salts is lost in the feces. The liver attempts to compensate by upregulating de novo but this compensatory capaci

When bile salt losses exceed hepatic synthetic capacity, the total bile salt pool shrinks. Intraluminal concentrations fall below the CMC, impairing micelle formation and reducing emulsification efficiency. The result is secondary steatorrhea, even though the pancreas and proximal enterocytes are functioning normally. Additionally, bile salts that reach the colon in excess stimulate colonic secretion, causing bile salt-induced diarrhea (cholerheic diarrhea), which may precede steatorrhea in patients with limited ileal resections.

**CLINICAL CORRELATION**

**SHORT BOWEL SYNDROME**

Short bowel syndrome follows extensive intestinal resection — most commonly for Crohn's disease, mesenteric ischemia, or trauma — and illustrates how multiple fat absorption defects can converge in a single patient. Resection of the terminal ileum disrupts ASBT-mediated bile salt reabsorption, depleting the enterohepatic pool below the critical micellar concentration and impairing micelle formation. Jejunal resection directly reduces absorptive surface area, while loss of the ileal brake (through reduced GLP-1 and PYY secretion) accelerates transit and shortens the contact time between micelles and the absorptive epithelium.

The severity of malabsorption depends on the length and location of resected bowel. Management requires high-calorie enteral nutrition, fat-soluble vitamin and B12 supplementation, and in severe cases, total parenteral nutrition.

**ACTIVE LEARNING QUESTION**

**A patient with Crohn's disease undergoes terminal ileal resection. Which finding would most likely be observed on stool analysis?**

- **A.** ○ Increased bile salt synthesis
- **B.** ○ Steatorrhea from reduced micelle formation
- **C.** ○ Pancreatic lipase deficiency
- **D.** ○ Enhanced cholesterol absorption
- **E.** ○ Portal hypertension

Explanation (requires correct answer)

**CASE CONNECTION**

BACK TO INTRODUCTION ↑

Thinking back to MS, what processes might account for his inability to absorb dietary fat? How would you distinguish between luminal and absorptive causes of fat malabsorption?

MS's bulky, oily stools are classic for **steatorrhea**, and the brick's framework points toward two diagnostic categories. **Luminal causes** include pancreatic exocrine insufficiency (cystic fibrosis, chronic pancreatitis) — confirmed by reduced fecal elasta function (Crohn's disease, ileal resection) or biliary obstruction. **Absorptive causes** include celiac disease — diagnosed by anti–tissue transglutaminase IgA serology and small bowel biopsy showing villous atrophy — and rare chylomicron assembly defects like abetalipoproteinemia, which would also produce markedly reduced circulating LDL and characteristic acanthocytes on peripheral smear.

Given MS's age, the absence of surgical or alcohol history, and the isolated GI presentation, celiac disease and cystic fibrosis are reasonable initial considerations alongside a chronic inflammatory ileal process. Across most of these conditions, **MCT supplementation** offers a common nutritional bridge: short- and medium-chain fatty acids bypass the lipase–bile salt–chylomicron pathway entirely, preserving caloric intake while the underlying cause is identified and treated.

## Summary

- Bile salts are amphipathic molecules that emulsify dietary fat droplets and form mixed micelles above the critical micellar concentration (CMC), solubilizing FFAs and 2-MGs for delivery to enterocytes; deficiency causes steatorrhea and fat-soluble vitamin deficiencies.
- Pancreatic lipase cleaves TGs at the sn-1 and sn-3 positions to yield FFAs and 2-MG; colipase is an essential cofactor that anchors lipase to the lipid droplet surface, overcoming displacement by bile salts.
- Lipid products enter enterocytes via passive diffusion and transporters (FATP4, CD36), are re-esterified to TGs via the monoglyceride pathway in the SER, and are assembled with apoB-48 into chylomicrons in the Golgi.
- Chylomicrons are too large to enter intestinal capillaries and instead enter lacteals, travelin circulation, bypassing first-pass hepatic metabolism; carbohydrates and amino acids travel via the portal vein directly to the liver.
- The enterohepatic circulation (EHC) recycles approximately 95% of bile salts via ASBT-mediated active reabsorption in the terminal ileum, allowing a small bile salt pool to support large daily digestive requirements.
- Ileal dysfunction (e.g., Crohn's disease, resection) disrupts bile salt reabsorption, depletes the bile salt pool below the CMC, and causes secondary fat malabsorption and steatorrhea even when pancreatic and enterocyte function are intact.

## Review Questions

**1. What is the main role of bile salts in fat digestion?**

- **A.** ○ Hydrolyze triglycerides directly
- **B.** ○ Emulsify fats and form micelles
- **C.** ○ Act as digestive enzymes
- **D.** ○ Directly transport lipids
- **E.** ○ Activate pancreatic enzymes

Explanation (requires correct answer)

**2. Which molecule restores pancreatic lipase activity displaced by bile salts?**

- **A.** ○ Cholecystokinin
- **B.** ○ Colipase
- **C.** ○ Phospholipase A2
- **D.** ○ Enteropeptidase
- **E.** ○ Procolipase

Explanation (requires correct answer)

**3. Which apolipoprotein is essential for chylomicron assembly?**

- **A.** ○ Apolipoprotein A-I
- **B.** ○ Apolipoprotein A-IV
- **C.** ○ Apolipoprotein B-48
- **D.** ○ Apolipoprotein E
- **E.** ○ Apolipoprotein C-II

Explanation (requires

**4. Why do chylomicrons enter lymphatics instead of intestinal capillaries?**

- **A.** ○ They are water-soluble
- **B.** ○ ApoB-48 directs them
- **C.** ○ Large size prevents capillary entry
- **D.** ○ Portal vein pressure excludes them
- **E.** ○ They carry fat-soluble vitamins

Explanation (requires correct answer)

**5. What is a likely effect of ileal resection on bile salt cycling?**

- **A.** ○ Increased bile salt synthesis
- **B.** ○ Steatorrhea from reduced micelles
- **C.** ○ Pancreatic lipase deficiency
- **D.** ○ Enhanced cholesterol absorption
- **E.** ○ Portal hypertension

Explanation (requires correct answer)

**How would you rate this Brick?**

## Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☐ **Predict how bile salt deficiency impairs lipid solubilization, and describe the amphipathic basis for emulsification and micelle formation.**
- 02 ☐ **Analyze the cooperative roles of pancreatic lipase and colipase in triglyceride hydrolysis and predict the absorptive consequences of lipase deficiency.**
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**Objective Confidence: 0/5**